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Started	Tuesday, 20 October 2026, 6:53 PM
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Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It slows down the process
b. It makes it easier to select items based on their relationship to the rest of the items
c. It complicates the selection of items
d. It makes the data unsorted

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **2**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Being the most efficient sorting algorithm
b. Using the divide-and-conquer approach
c. Bubbling up the largest element to its correct position with each pass
d. Sorting data in a non-sequential manner

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **3**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. Each part is searched for a specific element
b. Each part is sorted independently
c. The parts are ignored
d. The parts are merged without sorting

The correct answer is: Each part is sorted independently

Question **4**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Greedy method
b. Dynamic programming
c. Divide and conquer
d. Backtracking

The correct answer is:
Divide and conquer

Question **5**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are discarded
b. They are combined to form a single sorted list
c. They are split again into smaller sublists
d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Quick Sort
b. Bubble Sort
c. Heap Sort
d. Merge Sort

The correct answer is: Bubble Sort

Question **7**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It sorts data out of the box efficiently
- b. It is less efficient than custom sorting algorithms
- c. It only works with integer arrays
- d. It requires external libraries

The correct answer is: It sorts data out of the box efficiently

Question **8**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Bubble Sort
- b. Insertion Sort
- c. Merge Sort
- d. Quick Sort

The correct answer is: Merge Sort

Question **9**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Merge Sort
- b. Linear Search
- c. Bubble Sort
- d. Quick Sort

The correct answer is: Merge Sort

Question **10**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It has no effect on the analysis process
b. It complicates the analysis process
c. It makes the analysis slower
d. It allows for quicker identification of duplicates

The correct answer is: It allows for quicker identification of duplicates

Question **11**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Bubble Sort
b. Linear Search
c. Merge Sort
d. Binary Search

The correct answer is: Merge Sort

Question **12**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Merging
b. Searching
c. Sorting
d. Complexity

The correct answer is:
Complexity

Question **13**

Complete

Mark 0.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,80,90,150,200,300,400,470,70,120
- b. 40,70,80,90,120,150,200,300,400,470
- c. 200,470,80,150,40,90,300,400,70,120
- d. 80,150,200,470,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **14**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Makes code execution slower
- b. Decreases the efficiency of algorithms
- c. Provides a basis for other algorithms to work efficiently
- d. Makes data harder to manage

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **15**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It only works on small datasets
- b. It allows for efficient parallel processing and sorting of data
- c. It avoids the need for recursion
- d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data